Building a Common Control Plane for Multi-Vendor Fabrics

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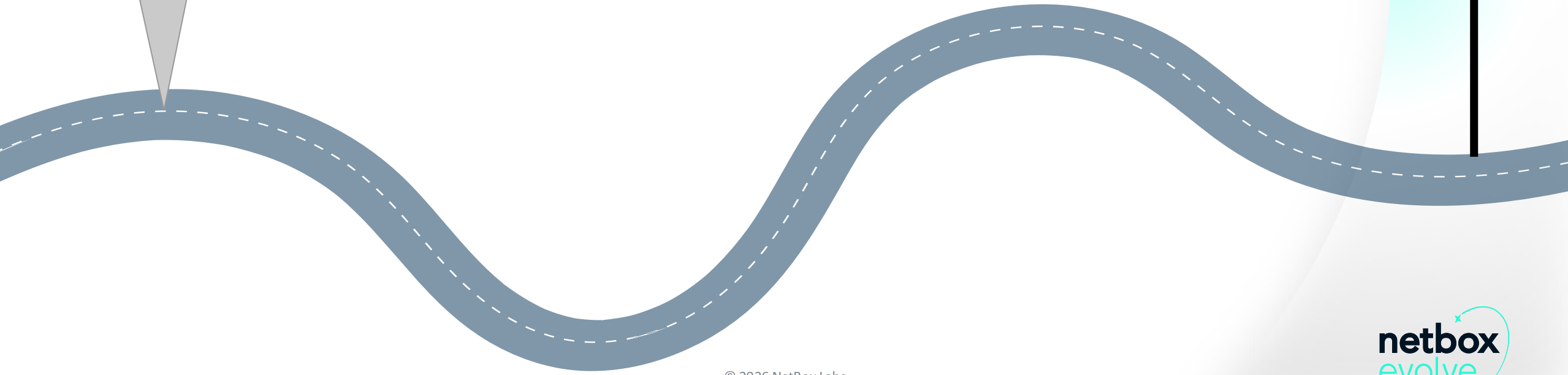
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How did we get here?





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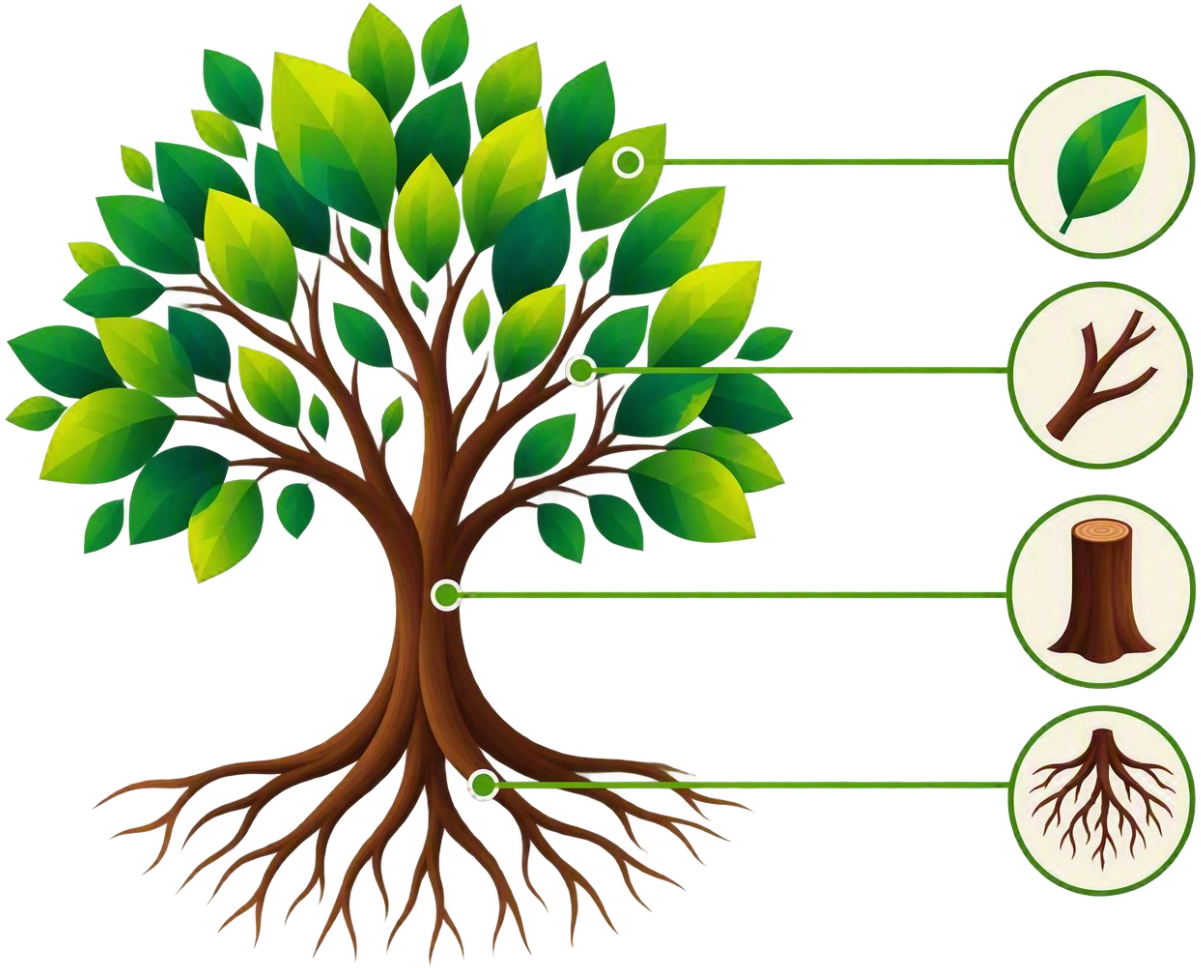
What do we mean by Common Control Plane?



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Common Characteristics

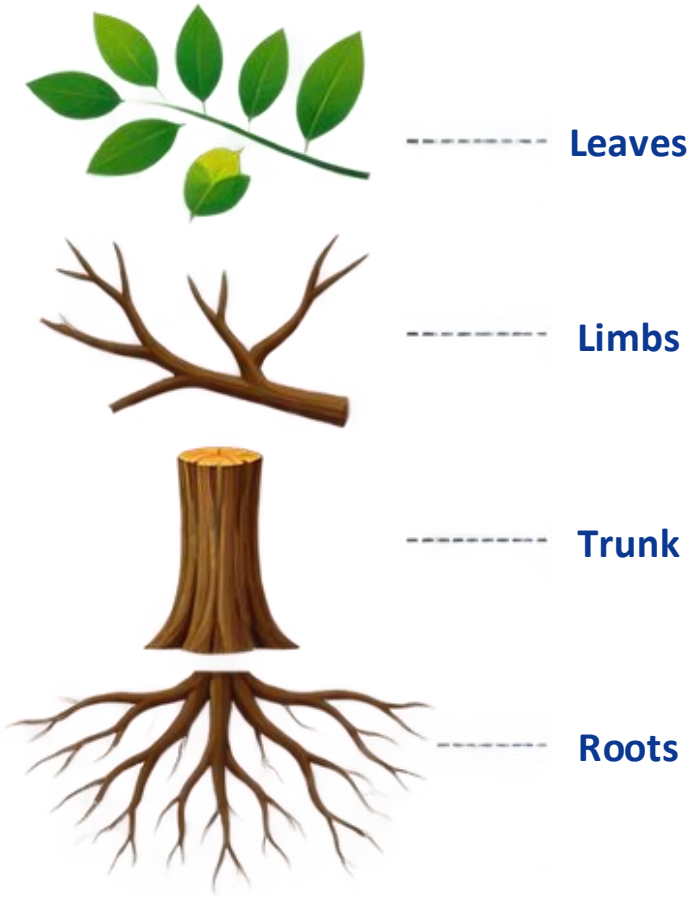
… our trees share fundamental components



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Dissaggregation

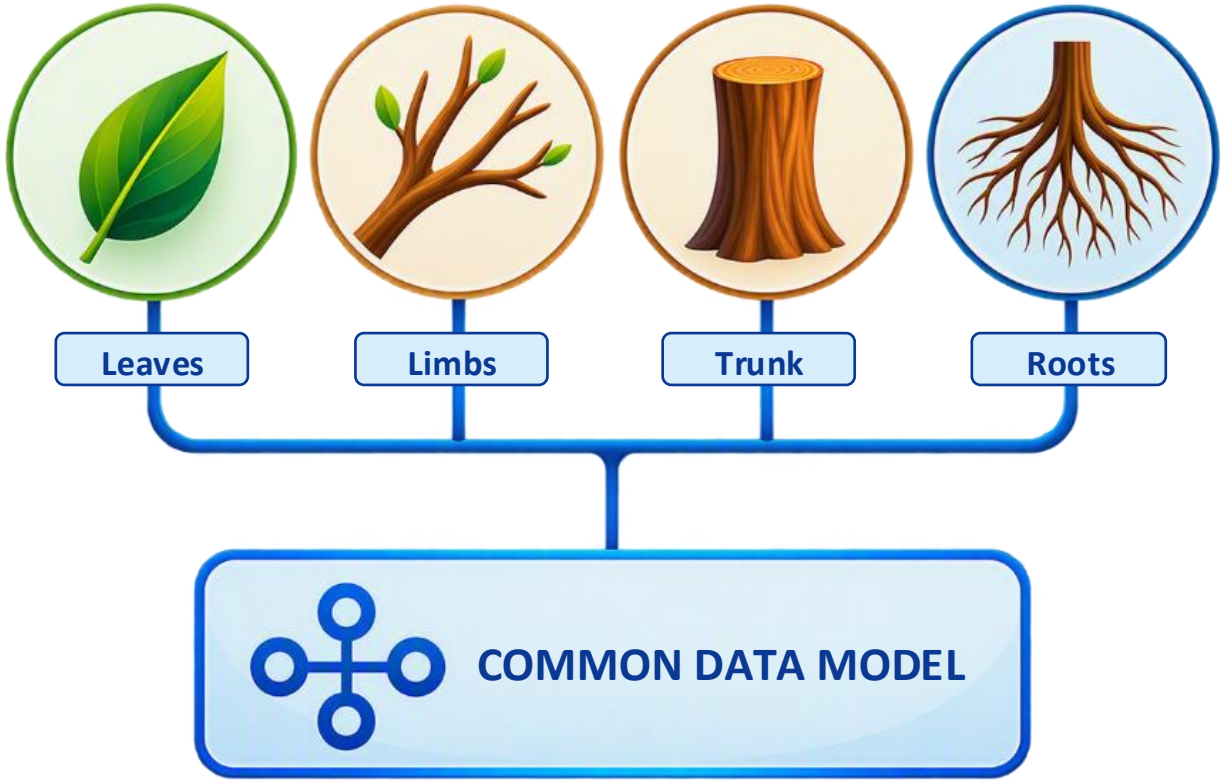
… Break the system into individual components



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Assemble the Data Model

... Use a common model to describe the components



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Apply our Data Model

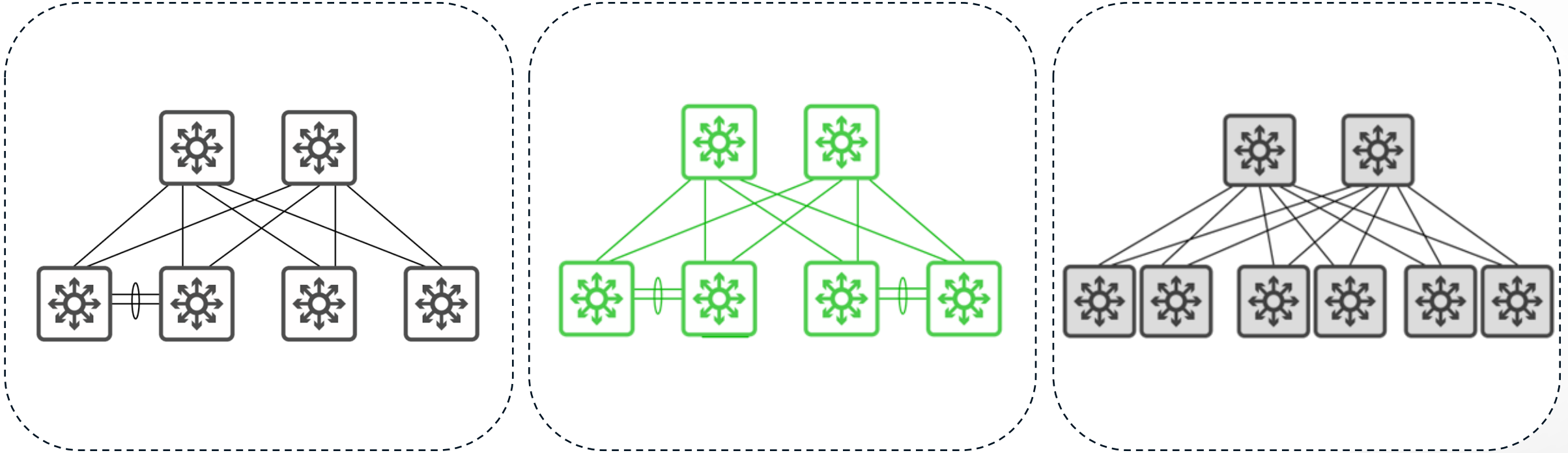
… Use our data model to describe unique systems



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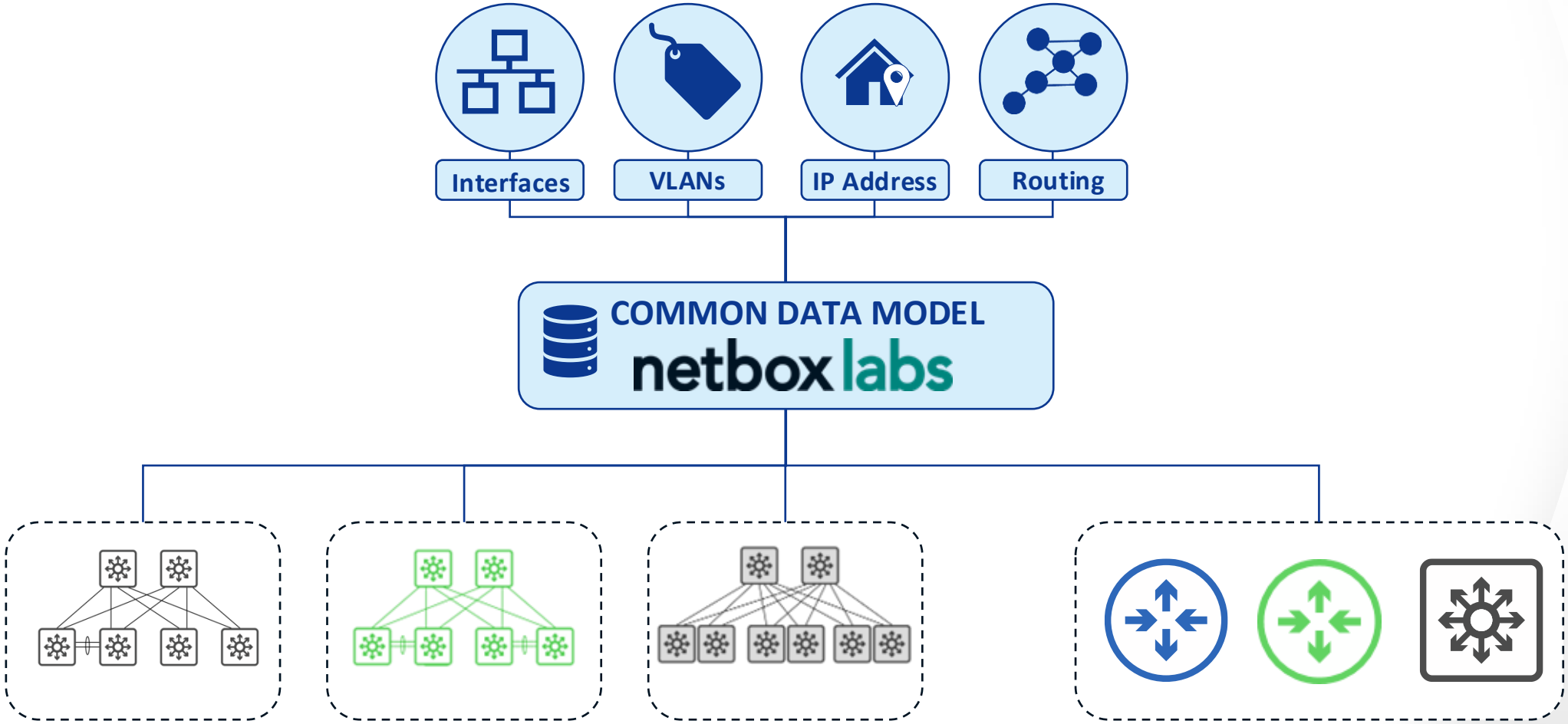
Our Network Fabrics as Trees

… Our network systems share common components

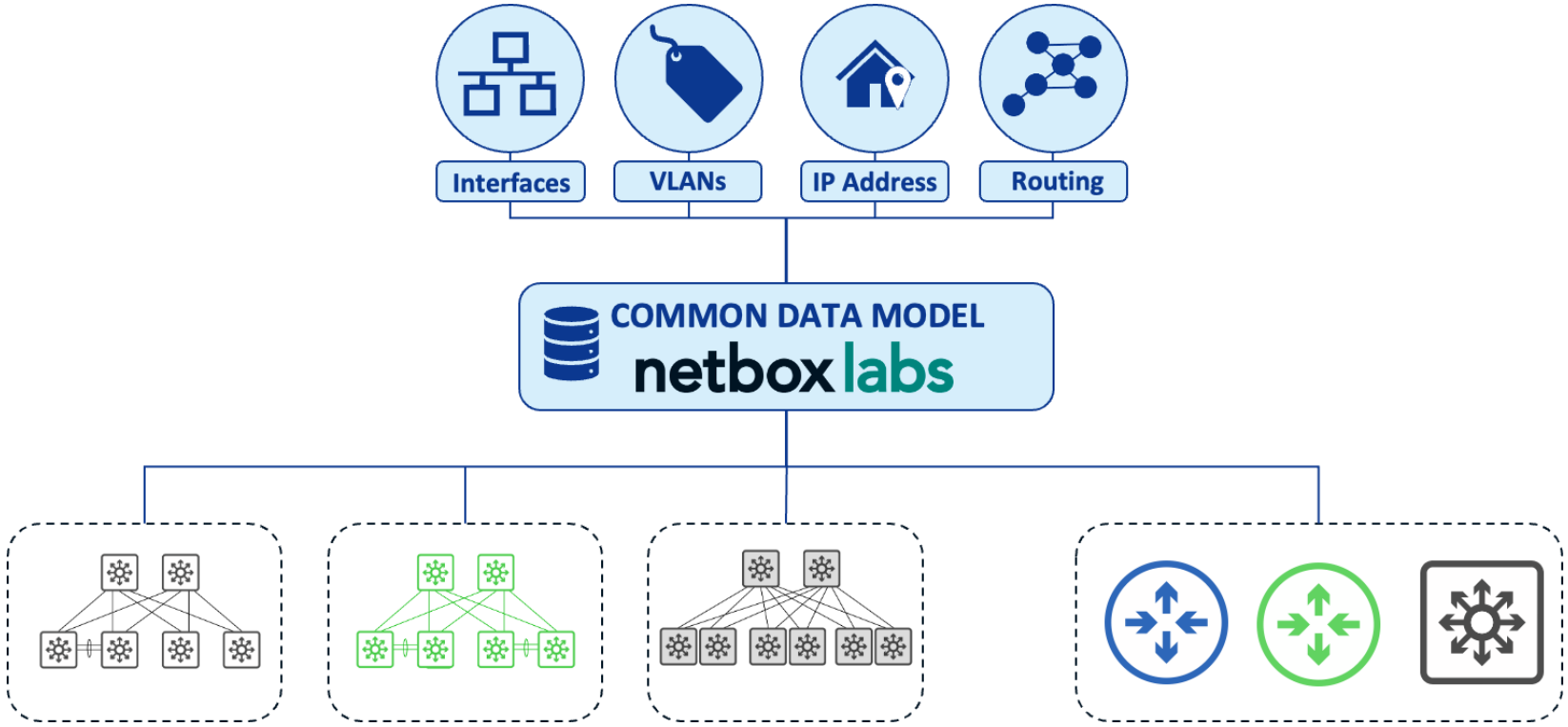


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…Described in a common data model



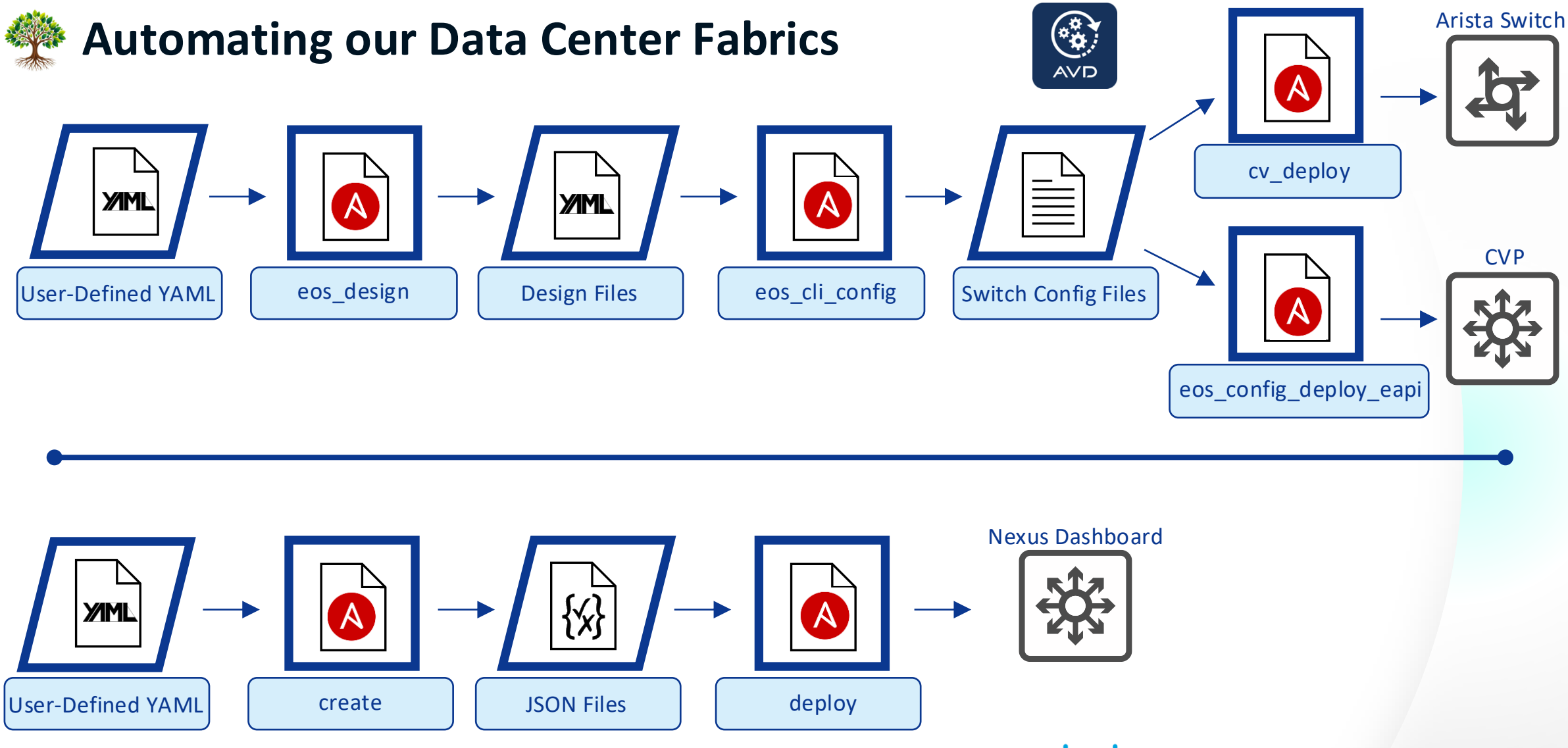
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Automating our Data Center Fabrics

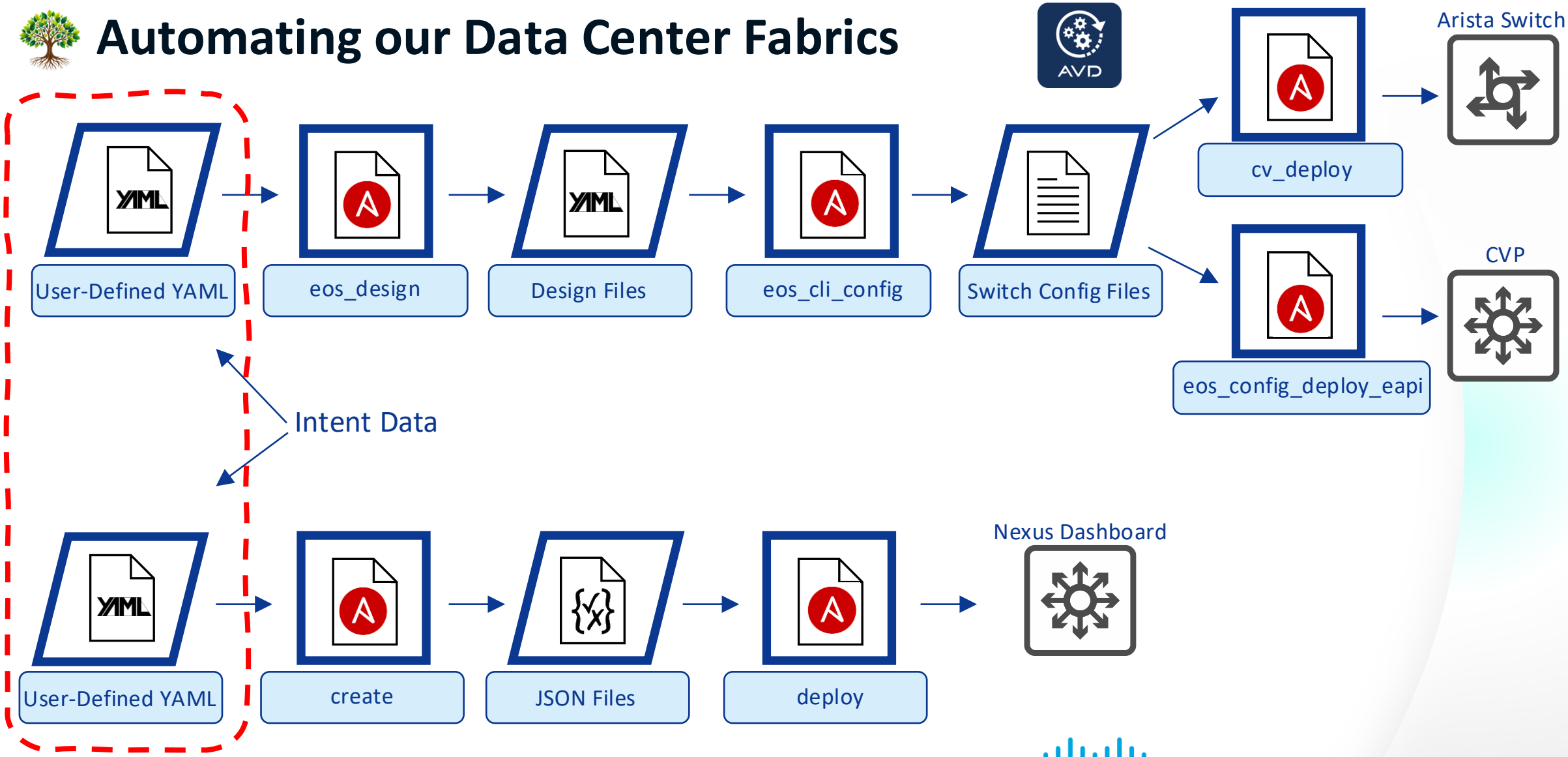


CISCO NAC

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Automating our Data Center Fabrics

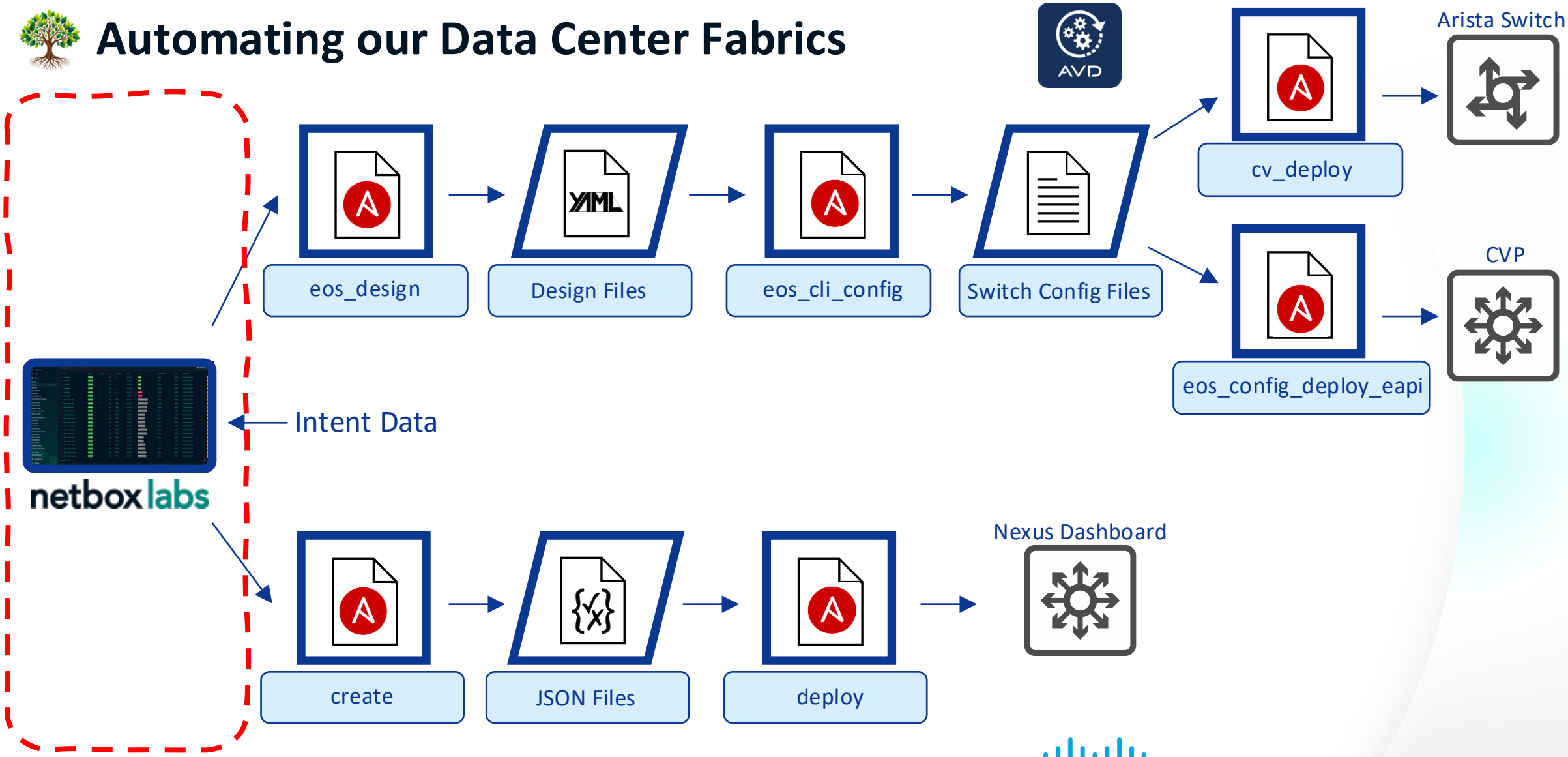


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Automating our Data Center Fabrics



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Why do this?

… Make life easier.

Unified View



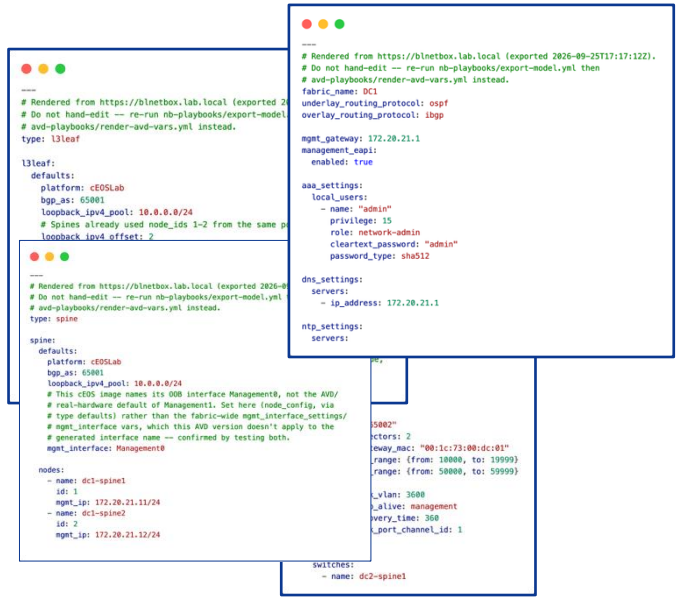
Multi-OEM Strategy

CISCO

ARISTA

JUNIPER NETWORKS

Ease of Management

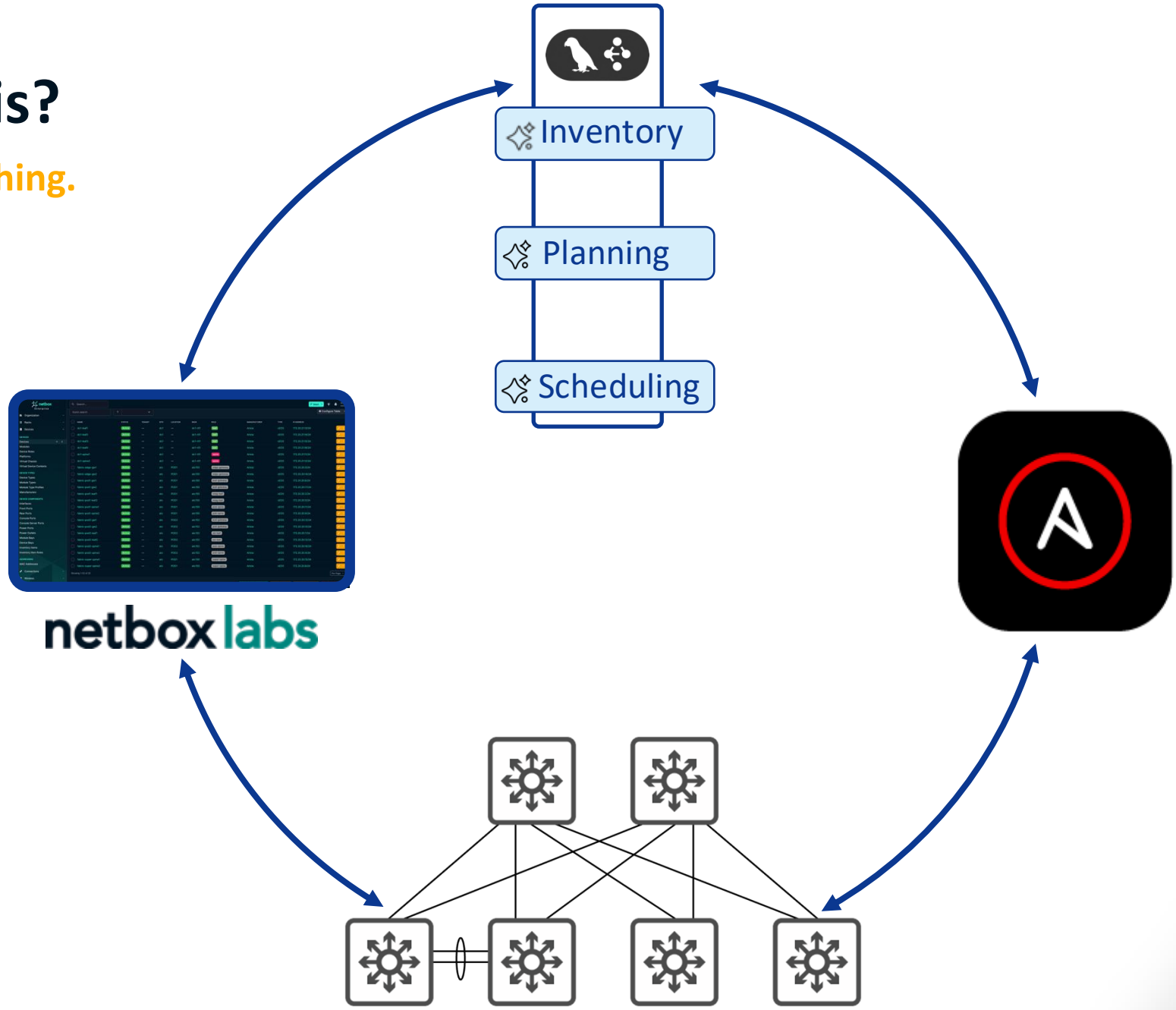


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Why do this?

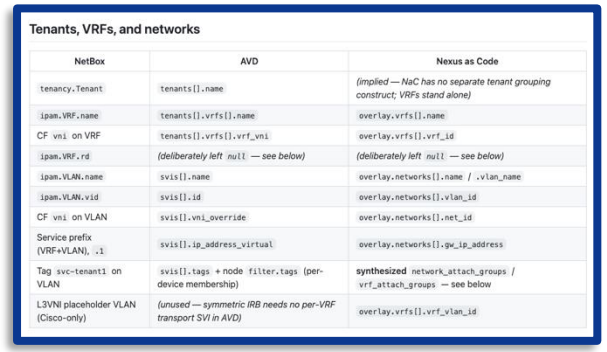
… data is everything.



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The Work

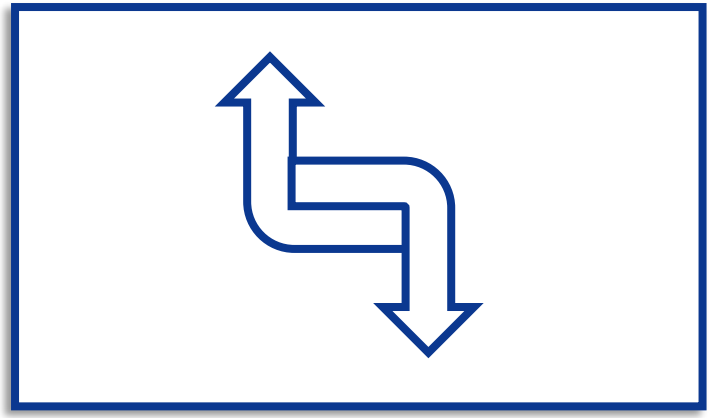
Step 1: Object Mapping



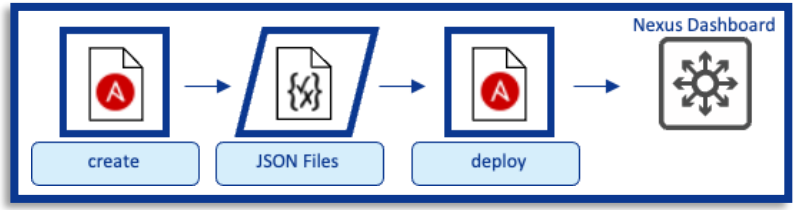
Tenants, VRFs, and networks

NetBox	AVD	Nexus as Code
`tenancy.Tenant`	`tenants[].name`	*(implied — NaC has no separate tenant grouping construct; VRFs stand alone)*
`ipam.VRF.name`	`tenants[].vrfs[].name`	`overlay.vrfs[].name`
CF `vni` on VRF	`tenants[].vrfs[].vrf_vni`	`overlay.vrfs[].vrf_id`
`ipam.VRF.rd`	*(deliberately left `null` — see below)*	*(deliberately left `null` — see below)*
`ipam.VLAN.name`	`svis[].name`	`overlay.networks[].name` / `.vlan_name`
`ipam.VLAN.vid`	`svis[].id`	`overlay.networks[].vlan_id`
CF `vni` on VLAN	`svis[].vni_override`	`overlay.networks[].net_id`
Service prefix (VRF+VLAN), `.1`	`svis[].ip_address_virtual`	`overlay.networks[].gw_ip_address`
Tag `svc-tenant1` on VLAN	`svis[].tags` + node `filter.tags` (per-device membership)	**synthesized** `network_attach_groups` / `vrf_attach_groups` — see below
L3VNI placeholder VLAN (Cisco-only)	*(unused — symmetric IRB needs no per-VRF transport SVI in AVD)*	`overlay.vrfs[].vrf_vlan_id`

Step 2: Render Objects



Step 3: Feed the OEM Solution



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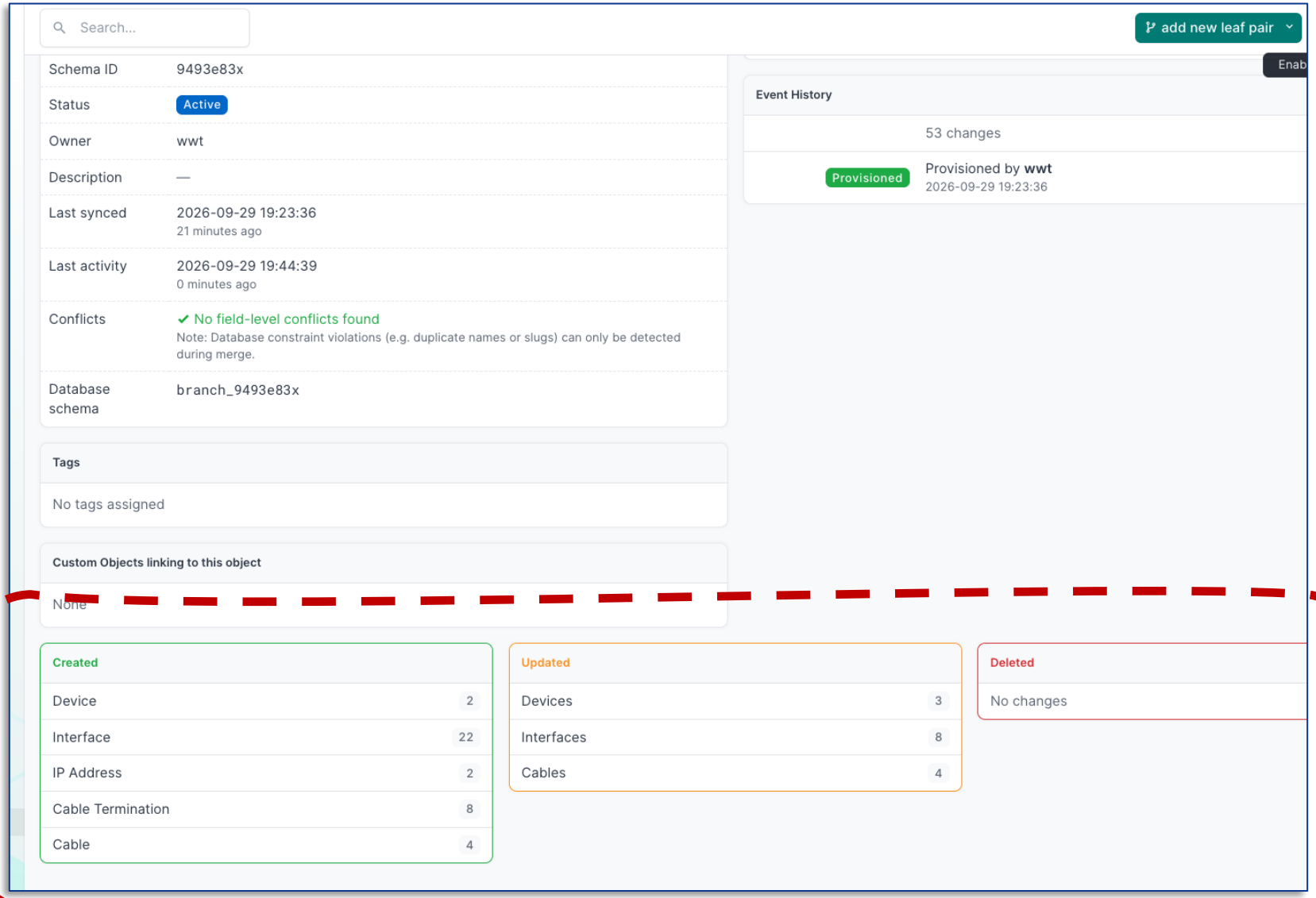
The Data Model in NetBox

NAME	STATUS	TENANT	SITE	LOCATION	RACK	ROLE	MANUFACTURER	TYPE	IP ADDRESS
dc1-leaf1	Active	—	dc1	—	dc1-r01	leaf	Arista	cEOS	172.20.21.13/24
dc1-leaf2	Active	—	dc1	—	dc1-r01	leaf	Arista	cEOS	172.20.21.14/24
dc1-leaf3	Active	—	dc1	—	dc1-r01	leaf	Arista	cEOS	172.20.21.15/24
dc1-leaf4	Active	—	dc1	—	dc1-r01	leaf	Arista	cEOS	172.20.21.16/24
dc1-spine1	Active	—	dc1	—	dc1-r01	spine	Arista	cEOS	172.20.21.11/24
dc1-spine2	Active	—	dc1	—	dc1-r01	spine	Arista	cEOS	172.20.21.12/24
dc2-leaf1	Active	—	dc2	—	dc2-r01	leaf	Cisco	N9K-C93180YC-EX	172.20.22.13/24
dc2-leaf2	Active	—	dc2	—	dc2-r01	leaf	Cisco	N9K-C93180YC-EX	172.20.22.14/24
dc2-leaf3	Active	—	dc2	—	dc2-r01	leaf	Cisco	N9K-C93180YC-EX	172.20.22.15/24
dc2-leaf4	Active	—	dc2	—	dc2-r01	leaf	Cisco	N9K-C93180YC-EX	172.20.22.16/24
dc2-spine1	Active	—	dc2	—	dc2-r01	spine	Cisco	N9K-C93180YC-EX	172.20.22.11/24

- Native Objects
- Custom Fields
- Custom Objects
- Config Contexts

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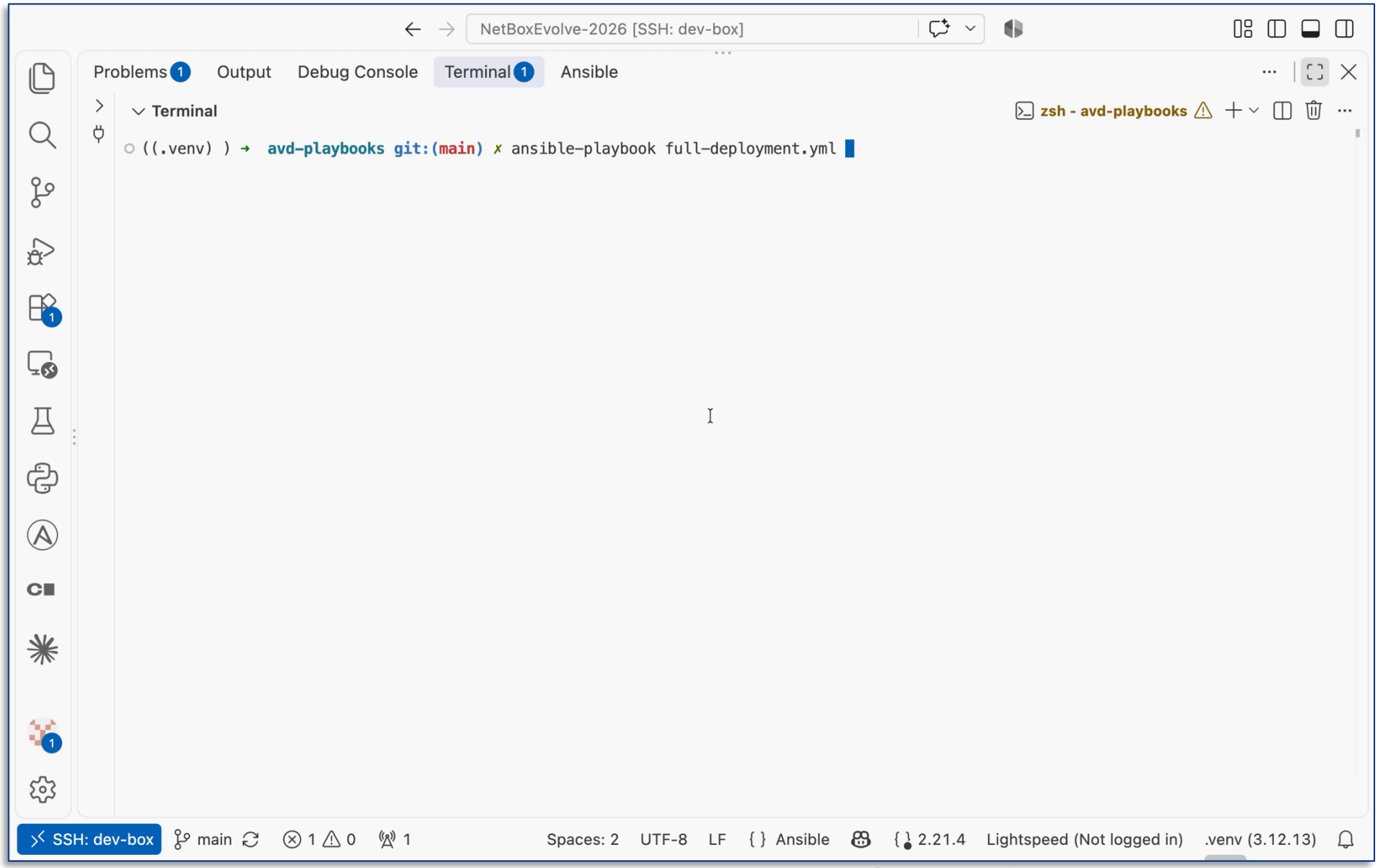
Staging Our Changes



Adding two new leaf switches in our “add new leaf pair branch”

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Deploy to Fabric



Merge Branch and Run Playbooks

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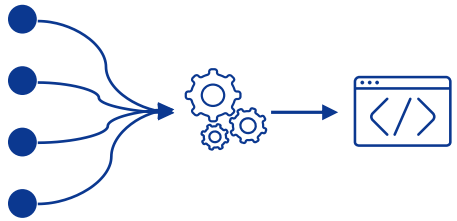
Now What?

... Start with your data

Scattered systems of record scale inconsistencies



Consolidated intent, enforced by automation



Trustworthy data enables trustworthy outcomes



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Thanks!

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